

Evaluating Expressions with RadicalsQuestion Bank - Additional 4.05 **ANSWER KEY**

*Show your work. Circle your final answer.***For questions 1 through 6, evaluate.**

1. $\sqrt{64}$

8

2. $\sqrt[3]{8}$

2

3. $\sqrt[3]{-125}$

-5*An odd root keeps the sign of its radicand.*

4. $\sqrt{\frac{9}{49}}$

 $\frac{3}{7}$ *Top and bottom each get the root.*

5. $\sqrt{4^2 + 33}$

7 $16 + 33 = 49.$

6. $\sqrt[3]{5^2 + 100}$

5 $25 + 100 = 125.$

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a. $\sqrt{144} \cdot 3^{-2}$

$$\frac{4}{3}$$

$$12 \cdot \frac{1}{9}$$

b. $\left(-\frac{1}{4}\right)^2 + \sqrt{3^2 + 27}$

$$\frac{97}{16}$$

$$\frac{1}{16} + 6$$

c. $\sqrt[3]{-27} - 2^{-1}$

$$-\frac{7}{2}$$

$$-3 - \frac{1}{2}$$

8. Evaluate each expression.

a. $\sqrt[3]{64} - 1.5(\sqrt{7^2 - 13})$

$$-5$$

$$4 - 1.5(6)$$

b. $\left(-\frac{1}{2}\right)^3 - \sqrt[3]{4^2 + 11}$

$$-\frac{25}{8}$$

$$-\frac{1}{8} - 3$$

c. $\sqrt{6^2 + 13} \cdot 2^{-3}$

$$\frac{7}{8}$$

$$7 \cdot \frac{1}{8}$$

Working backwards.

9. Find the value of n that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a. $\sqrt{n} = 15$

$n = 225$

b. $\sqrt[3]{n} = -2$

$n = -8$

c. $\sqrt{n-5} = 11$

$n = 126$

$n - 5 = 121$.

Reading someone else's work.

10. Deshawn writes $\sqrt{100-36} = \sqrt{100} - \sqrt{36} = 4$. His answer happens to be wrong by a small amount. Evaluate the expression correctly, and explain why his method cannot be trusted even when it lands close.

$\sqrt{100-36} = \sqrt{64} = 8$. *The bar groups the subtraction, so it is finished before the root is taken; the root of a difference is not the difference of the roots, and how close his answer lands is an accident of the numbers.*

Ask him to try $\sqrt{25-9}$: his method gives 2 and the value is 4.

11. Which is the value of $\sqrt{8^2 + 36}$?

☒ A. 10

☐ B. 100

☐ C. 50

☐ D. 14

A

12. Select ALL of the expressions whose value is 4.

☒ A. $\sqrt{16}$

☒ B. $\sqrt[3]{64}$

☒ C. $\sqrt{5^2 - 9}$

☒ D. $\sqrt[3]{3^3 + 37}$

☐ E. $\sqrt[3]{-64}$

☐ F. $\sqrt{2^3 + 17}$

A, B, C and D